Assignment 4 - strings

100%

10.10.2023

Poengsum frakoblet modus: **100%**

Legg til kommentar

Anonym vurdering: **nei**

15.12.2023

Detaljer

Purpose

Note: Starting on this assignment before the *strings* lecture will be tough. If you are following the normal course progression I would read the lecture notes to get an idea what this is about, then wait until after the string lecture to work on this assignment.

This week we will continue with our new building block, functions, and add strings to the mix. You will both use functions from the standard library and write some of your own.

4.1

Folder name: ikt101g23h/assignments/solutions/assignment_4_1

Write an application that asks the user for a word and:

- Tells how many letters are in the word
- Tells if the word is a palindrome
- Prints the word reversed

Notes:
Do not use the *strrev* function. It is not available on Linux and your code will not pass Bamboo tests. Write your own.

Requirements:
Checking if the word is a palindrome must be done in a function returning bool called *is_palindrome*. *bool* is a type from *stdbool.h* that can only have the values *true* or *false*. See **this. (https://en.wikibooks.org/wiki/C_Programming/stdbool.h)**
Reversing the string must be done in function returning nothing (void) called *string_reverse*.

Expected output:

- The word contains *5* letters
- The word is a palindrome / The word is not a palindrome
- The word reversed is '*olleh*'

4.2

Folder name: ikt101g23h/assignments/solutions/assignment_4_2

Write an application that asks for a word as input and:

- Prints the word in uppercase and lowercase
- Splits the word on the middle and prints the two parts with '-' (a dash) between

Requirements:

Converting the word to upper and lower case must be done in functions. These functions must return nothing (void) and be called: *string_upper*, *string_lower*.
Do not use *strlwr* or *strupr*, they are not standard and will not work on Bamboo.

Note: The input word length is always even.

Expected output (with the string you receive as input):

Input is: ABCDefgh

- The word in uppercase is 'ABCDEFGH'
- The word in lowercase is 'abcdefgh'
- The word split in two is 'ABCD - efgh'

4.3

Folder name: ikt101g23h/assignments/solutions/assignment_4_3

Write an application that asks for two words and:

- Tells if the words are equal or not
- Tells if one word is a substring of the other

Expected output:

- The words are equal **OR** The words are not equal
- Word 1 is a substring of word 2 **OR** Word 2 is a substring of word 1 **OR** No substrings found

4.4

Folder name: ikt101g23h/assignments/solutions/assignment_4_4

Write an application that asks for a string and:

Tells how many times each character of the alphabet appears in the string

Example input: aaaaabbccc
Expected output:
'a' : 5

'b' : 2
'c' : 3

Test locally

see *Local testing with Docker* under *Assignments*.

Test the solution on your own computer. You can fix your code and retest as many times as you want.

Deliver and test in Bamboo

See *Delivery and Bamboo testing* under *Assignments*.

Finally deliver and test the solution in Bamboo. This is what updates your score in Canvas which is the score I can see. You can test as many times as you want here as well up until the final deadline.